B.TECH DEGREE I SEMESTER EXAMINATION DECEMBER 2019

EC/IT/CS 19-200-0105B BASIC ELECTRONICS ENGINEERING

(2019 Scheme)

Time: 3 Hours Total Marks:72
Maximum marks: 60

PART A

(Answer all the questions) Marks (3*8=24)

1.

a. Explain classification of material based on energy bands?
b. Write diode equation and explain each parameter.
c. The input voltage applied to the primary of a 5:1step down transformer of a bridge rectifier is 230 V ,50Hz.If the load resistance IS 800Ω and assuming diodes are ideal find DC power delivered to the load and rectification efficiency
d. Write notes on effect of variation of temperature on forward voltage drop and dynamic resistance of a diode
e. Compare transistor configurations
f. For an **npn** transistor α=0.98 and I_C =10mA.Find the value of β,I_c and I_E .
g. Explain various types of ICs.
h. Explain the equation for determining power gain and power level changes when two power levels are measured and when two voltage levels are measured.

PART B Marks (12*4=48)

2.

a. Differentiate between intrinsic and extrinsic semiconductors. (3)
b. Define donor doping and acceptor doping with necessary diagram (5)
c. The reverse saturation current for a PN junction is 25nA at 25^0C.Calculate I_0 at temperature of 30^0C, 33^0C and 37^0C (4)

OR

3.

a. A sample of silicon is doped with 0.5×10^{14} acceptor atoms/cm^3 and 3×10^{14} donor atoms/cm^3. Determine the resultant density of free electrons and holes in the sample. (4)
b. Calculate the drift velocity for electrons and holes in a 1mm length of silicon at 27^0C when the terminal voltage is 10V (4)
c. Explain the formation of depletion region in pn junction. (4)

4.

a. Explain the working of a full wave bridge rectifier with capacitive filter. (10)
b. What is the purpose of filter? Explain any two filter (2)

OR

5.

a. Explain the forward and reverse characteristics of pn junction diode. (5)
b. What is a Zener breakdown? Explain the role of Zener diode in voltage regulation. (4)
c. Write short note on seven segment display. (3)

6.

a. Compare different transistor configurations. (3) .
b. With help of neat block diagram explain CRO (5)
c. Discuss how a BJT is tested by means of an ohmmeter and a digital multimeter. (4)

OR

7.

a. With help of neat block diagram explain power supply. (4)
b. Sketch typical BJT common emitter input & output characteristics(4)
c. How transistor works as an amplifier? (4)

8. a. Explain IC fabrication using silicon planar technology. (8)
b. Draw the sketches to show the construction of alloy diode and diffuced diodes

OR

9.

a. Explain the process of preparing semiconductor material for device manufacturing (6)
b. Briefly discuss the requirements for constructing transistors with good current gain and high frequency response(4)

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